

Lecture 14

EE-215 Electronic Devices and Circuits

Asst Prof Muhammad Anis Ch

Biassing in BJT Amplifier Circuits

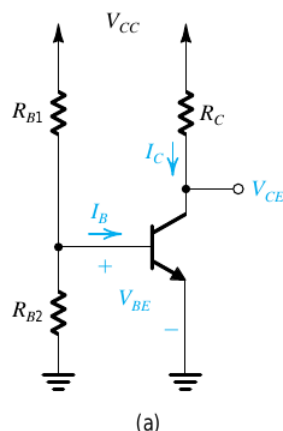
- we have already determined
 - In the design of a BJT amplifier circuit,
 - it is essential to establish an appropriate dc operating point (or Q point or bias point) for the BJT
 - This step is known as biasing or bias design
 - An appropriate bias point or dc operating point is characterized by
 - a stable and predictable dc collector current I_C and
 - by a dc collector-to-emitter voltage V_{CE} that ensures operation in the active region for all expected input signal levels

Biassing by Fixing V_{BE} or Biassing by Fixing I_B

- Two obvious schemes for biasing the BJT are
 1. Biassing by fixing V_{BE}
 2. Biassing by fixing I_B
 - Both these schemes result in wide variations in I_C and hence in V_{CE}
 - and are therefore not recommended for practical amplifiers

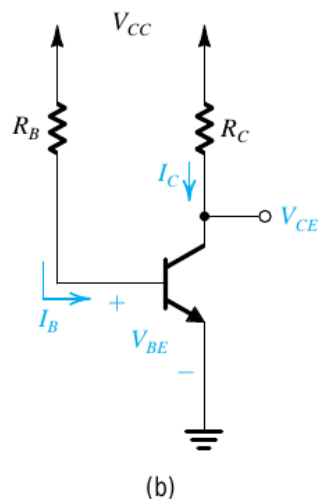
Biassing by Fixing V_{BE}

- in this case BJT is biased by fixing its V_{BE}
 - This voltage V_{BE} can be easily derived from the power-supply V_{CC}
 - by using an appropriate voltage divider
 - however, this way of biasing a BJT is not a viable approach
 - because the very sharp exponential relationship $i_C - v_{BE}$ means that any small and inevitable differences in V_{BE} from the desired value will result in large differences in I_C and in V_{CE}
 - $I_C = I_S e^{V_{BE}/V_T}$,
 - $V_{CC} = I_C R_C + V_{CE} \Rightarrow V_{CE} = V_{CC} - I_C R_C$



Biassing by Fixing I_B

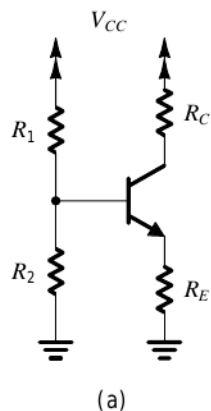
- in this case BJT is biased by establishing a constant current in the base, I_B
 - here by ohms' law
 - $I_B = \frac{V_{CC} - V_{BE}}{R_B} \approx \frac{V_{CC} - 0.7}{R_B} \because V_{BE} \approx 0.7V$
 - again this is not a recommended approach
 - because typically large variations in the value of β exist among units of the same device type
 - this variation in β will result in large variation in I_C (as $I_C = \beta I_B$)
 - and hence in V_{CE} (as $V_{CE} = V_{CC} - I_C R_C$)



- thus Biasing by fixing V_{BE} and Biasing by fixing I_B
 - result in wide variations in I_C and V_{CE}
 - and are therefore not viable options for practical BJT amplifiers
 - the following are the practical ways of biasing a BJT, which would result in stable bias point
 1. The classical discrete-circuit bias arrangement
 2. A two-power-supply version of the classical bias arrangement
 3. Biasing using a collector-to-base feedback resistor
 4. Biasing using a constant-current source

The Classical Discrete-circuit Bias Arrangement

- An effective (and most commonly used) biasing technique for discrete BJT circuits,
 - consists of applying the base of the BJT with a fraction of the supply voltage V_{CC} through a voltage divider R_1, R_2
 - and connecting a resistance R_E in the emitter lead



- the voltage divider circuit can be replaced by its Thevenin equivalent

◦ here $V_{TH} = V_{BB} = \frac{R_2}{R_2 + R_1} V_{CC}$

- and

$$R_{TH} = R_B = R_1 \parallel R_2 = \left(\frac{1}{R_1} + \frac{1}{R_2} \right)^{-1} = \left(\frac{R_2 + R_1}{R_1 R_2} \right)^{-1} = \frac{R_1 R_2}{R_1 + R_2}$$

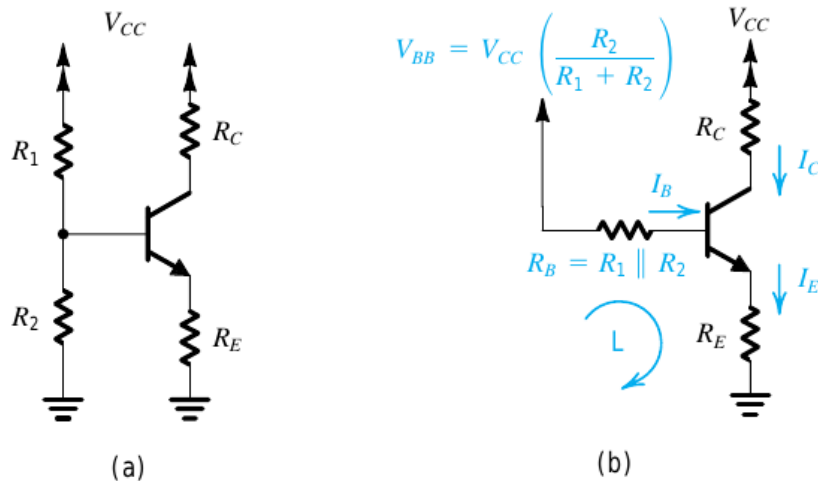


Figure 6.60 Classical biasing for BJTs using a single power supply: **(a)** circuit; **(b)** circuit with the voltage divider supplying the base replaced with its Thévenin equivalent.

- By KVL

◦ $V_{BB} = I_B R_B + V_{BE} + I_E R_E$

- As $I_B = \frac{I_E}{\beta + 1}$

- $\Rightarrow V_{BB} = \frac{I_E}{\beta + 1} R_B + V_{BE} + I_E R_E$

- or $V_{BB} - V_{BE} = I_E \left(\frac{R_B}{\beta + 1} + R_E \right)$

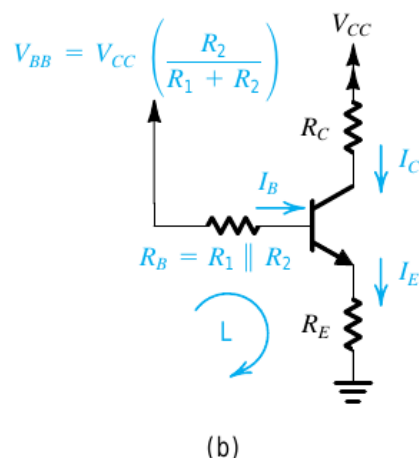
- $I_E = \frac{V_{BB} - V_{BE}}{\left(\frac{R_B}{\beta + 1} + R_E \right)}$

- As $I_C = \alpha I_E$ and as α varies very little.

- e.g. $\beta = 150 \Rightarrow \alpha = \frac{150}{151} = 0.9934$ and

- $\beta = 50 \Rightarrow \alpha = \frac{50}{51} = 0.9804$

- $\Rightarrow$ a stable I_E will result in equally stable I_C and vice versa



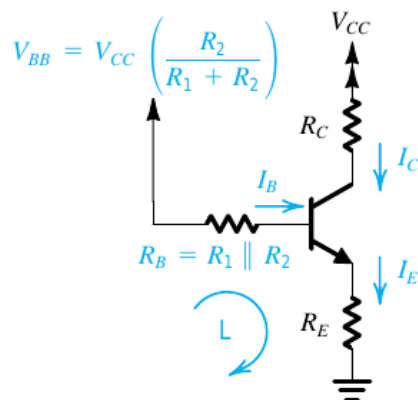
- thus $I_E = \frac{V_{BB} - V_{BE}}{\left(\frac{R_B}{\beta + 1} + R_E\right)}$

- $\Rightarrow$ to make I_E (and hence I_C) insensitive to temperature and β variations,

- we design the circuit to satisfy the following two constraints

1. $V_{BB} > V_{BE}$ (As V_{BE} changes with temperature)

2. and $R_E > \frac{R_B}{\beta + 1}$



(b)

- Condition 1 ($V_{BB} > V_{BE}$) ensures that the numerator of the expression for I_E stays the same, despite small changes in V_{BE}

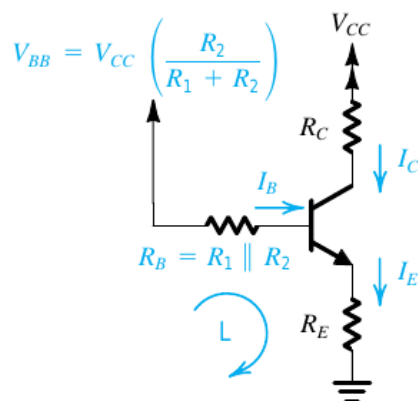
- however there is limit, on how large V_{BB} can be made.

- for a given value of V_{CC} , the higher the value of V_{BB} , the lower will be the sum of voltages across R_C and the V_{CE}

- to obtain high gain and large signal swing (+ve signal swing, avoid cutoff), we need to have large value of $I_C R_C$

- we also require large V_{CE} to ensure large signal swing (-ve signal swing, avoid saturation)

- $I_E = \frac{V_{BB} - V_{BE}}{\left(\frac{R_B}{\beta + 1} + R_E\right)}$



(b)

- Condition 1 ($V_{BB} > V_{BE}$)

- thus we have a conflicting set of requirements and the solution must be a trade-off

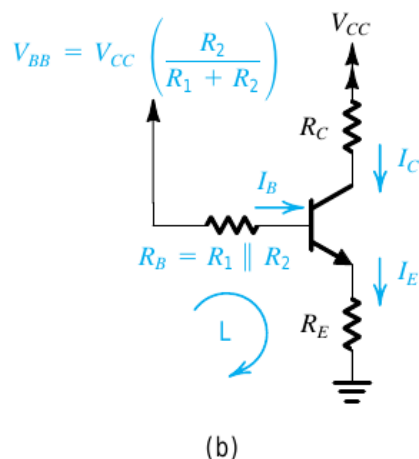
- As a rule of thumb, we can design for $V_{BB} \approx \frac{1}{3}V_{CC}$, $V_{CB} \approx \frac{1}{3}V_{CC}$ and $I_C R_C \approx \frac{1}{3}V_{CC}$

- Note that here we have ignored the voltage drop across the resistor R_B

- to include this drop, we can use $V_{BB} - I_B R_B = \frac{1}{3}V_{CC}$,
 $V_{CB} = \frac{1}{3}V_{CC}$ and $I_C R_C = \frac{1}{3}V_{CC}$

- (As the KVL equation would be
 $V_{CC} = I_C R_C + V_{CB} - I_B R_B + V_{BB}$)

- $$I_E = \frac{V_{BB} - V_{BE}}{\left(\frac{R_B}{\beta + 1} + R_E\right)}$$



- Also by KVL, $V_{BB} = I_B R_B + V_B \Rightarrow V_B = V_{BB} - I_B R_B$

- thus the rule of thumb is $V_B = \frac{1}{3}V_{CC}$, $V_{CB} = \frac{1}{3}V_{CC}$ and $I_C R_C = \frac{1}{3}V_{CC}$

- Condition 1 ($V_{BB} > V_{BE}$)

- thus the rule of thumb is $V_B = \frac{1}{3}V_{CC}$, $V_{CB} = \frac{1}{3}V_{CC}$ and $I_C R_C = \frac{1}{3}V_{CC}$

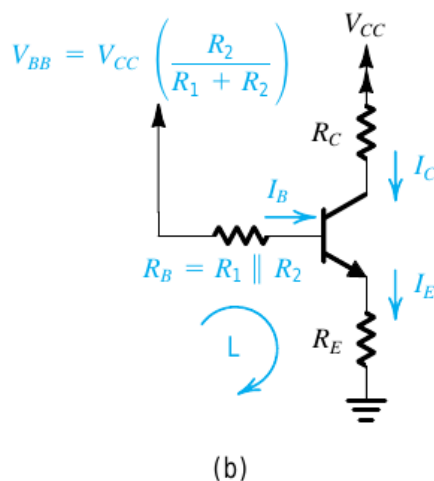
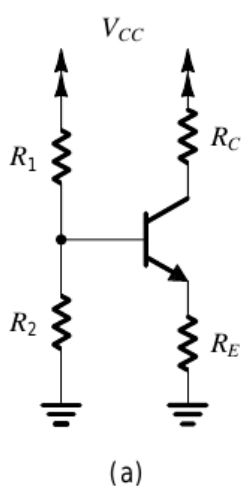


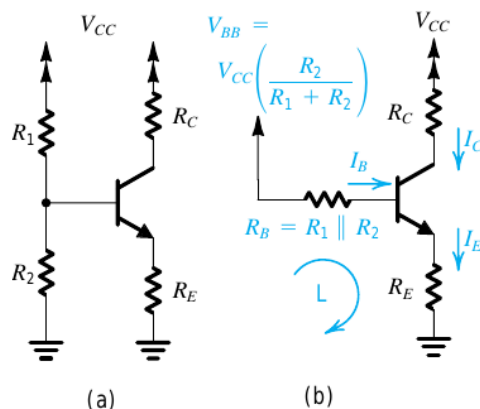
Figure 6.60 Classical biasing for BJTs using a single power supply: (a) circuit; (b) circuit with the voltage divider supplying the base replaced with its Thévenin equivalent.

- Condition 2 ($R_E > \frac{R_B}{\beta + 1}$) makes I_E insensitive to variations in β
 - and could be satisfied by selecting R_B small

- As $R_B = R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2}$, R_B = small means low values for R_1 and R_2

■ $\Rightarrow$ the input resistance of the amplifier is lowered (if the input is coupled to the base, R_1 and R_2 will appear in parallel between the base and ground)

- $$I_E = \frac{V_{BB} - V_{BE}}{\left(\frac{R_B}{\beta + 1} + R_E\right)}$$



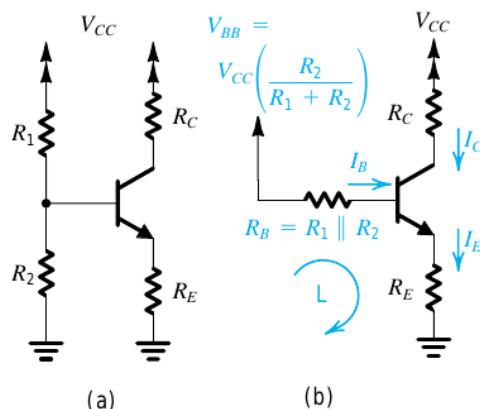
- Note that this condition ($R_E > \frac{R_B}{\beta + 1}$) $\Rightarrow R_B = R_1 \parallel R_2$ is very small i.e. the base voltage is independent of the value of the β and is determined solely by the voltage divider

- Condition 2 ($R_E > \frac{R_B}{\beta + 1}$)

- this condition can be satisfied by setting the current in the divider to be much larger than the base current

- typically one selects R_1 and R_2 such that their current is in the range of I_E to $0.1I_E$

- $$I_E = \frac{V_{BB} - V_{BE}}{\left(\frac{R_B}{\beta + 1} + R_E\right)}$$

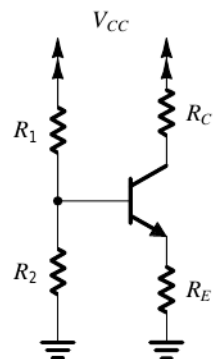


- Note that here the resistor R_E tends to stabilize the bias current I_E (and hence I_C)

- Consider that for some reason the emitter current increases.

- $\Rightarrow$ the voltage drop across R_E and hence V_E will increase
- As because of condition 2, V_B is determined primarily by voltage divider R_1, R_2 i.e. V_B will stay fixed
- $\Rightarrow V_{BE} = V_B - V_E$ has decreased

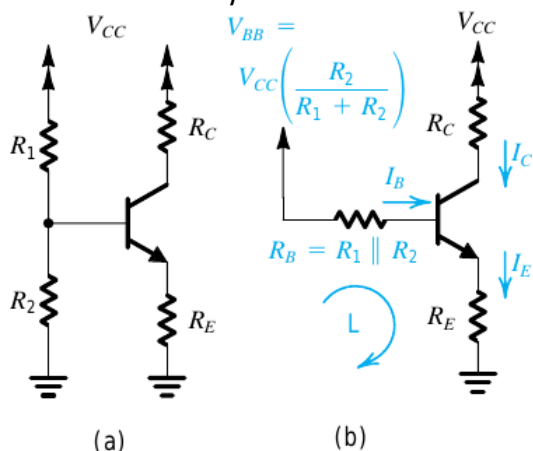
- this in turn will reduce the emitter (and collector) current
- thus R_E stabilizes the bias current



(a)

Example 6.20

- We wish to design the bias network of the amplifier in Fig. 6.60 to establish a current $I_E = 1mA$ using a power supply $V_{CC} = +12V$. The transistor is specified to have a nominal β value of 100.



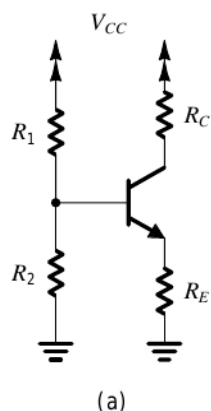
(a)

(b)

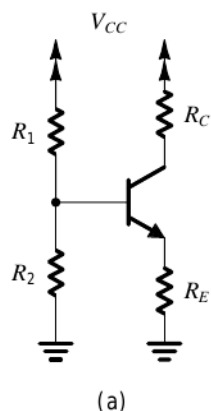
- here $I_E = 1mA$, $V_{CC} = 12V$, $\beta = 100$

$$\circ \Rightarrow \alpha = \frac{\beta}{\beta+1} = \frac{100}{101} = 0.9901$$

- $I_C = \alpha I_E = 0.9901 \times 1m = 0.9901mA$
- As a rule of thumb, this biasing circuit can be designed by setting
 - $V_B = \frac{1}{3}V_{CC}$, $V_{CB} = \frac{1}{3}V_{CC}$ and $I_C R_C = \frac{1}{3}V_{CC}$
 - here last two conditions ($V_{CB} = \frac{1}{3}V_{CC}$ and $I_C R_C = \frac{1}{3}V_{CC}$) will ensure enough signal swing
 - 1st condition ($V_B = \frac{1}{3}V_{CC}$) will ensure bias point stability
- $\Rightarrow V_B = 4V$, $V_{CB} = 4V$ and $I_C R_C = 4V$

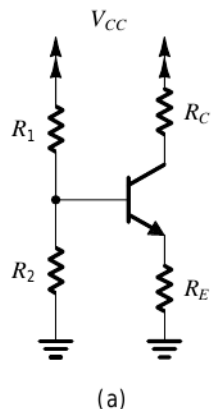


- $V_B = 4V$, $V_{CB} = 4V$ and $I_C R_C = 4V$
 - $V_{CB} = 4V \Rightarrow V_C - V_B = 4V$ or $V_C = V_B + 4 = 8V$
 - As $V_{BE} = 0.7V \Rightarrow V_B - V_E = 4 - V_E = 0.7$ or $V_E = 4 - 0.7 = 3.3V$
 - by ohms law
 - $I_E R_E = V_E = 3.3 \Rightarrow R_E = \frac{V_E}{I_E} = \frac{3.3}{1m} = 3.3k\Omega$
 - As $I_C R_C = 4V$ and $I_C = 0.9901mA$
 - $\Rightarrow R_C = \frac{4}{I_C} = \frac{4}{0.9901m} = 4.04k\Omega$
 - As the current through R_2 can be set to be in the range of I_E to $0.1I_E$

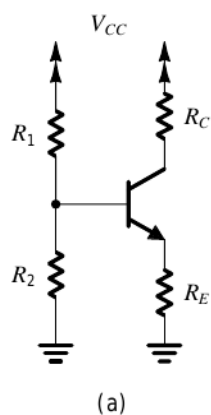


- Design 1: for $I_{R2} = 0.1I_E$
 - $I_{R2} = 0.1 \times 1m = 0.1mA$
 - by ohms law
 - $I_{R2} = \frac{V_B}{R_2} \Rightarrow R_2 = \frac{V_B}{I_{R2}} = \frac{4}{0.1m} = 40k\Omega$
 - Also by ohms law
 - $I_{R1} = \frac{V_{CC} - V_B}{R_1} = \frac{12 - 4}{R_1}$ or $R_1 = \frac{8}{I_{R1}}$
 - By KCL
 - $I_{R1} = I_{R2} + I_B = 0.1m + \frac{I_E}{\beta + 1} = 0.1m + \frac{1m}{101} = 0.1099mA$
 - $\Rightarrow R_1 = \frac{8}{I_{R1}} = \frac{8}{0.1099m} = 72.793k\Omega$
 - to confirm $I_E = \frac{V_{BB} - V_{BE}}{\left(\frac{R_B}{\beta + 1} + R_E\right)}$

$$\begin{aligned} \blacksquare \text{ here } V_{BB} &= \frac{V_{CC}R_2}{R_1+R_2} = \frac{12 \times 40k}{72.793k+40k} = \frac{480k}{112.793k} = 4.2556V, \\ R_B &= R_1 \parallel R_2 = 25.815k\Omega \\ \blacksquare I_E &= \frac{V_{BB}-V_{BE}}{\left(\frac{R_B}{\beta+1}+R_E\right)} = \frac{4.2556-0.7}{\left(\frac{25.815k}{101}+3.3k\right)} = 1mA \end{aligned}$$



- Design 2: for $I_{R2} = I_E$
 - $I_{R2} = 1m = 1mA$
 - by ohms law
 - $I_{R2} = \frac{V_B}{R_2} \Rightarrow R_2 = \frac{V_B}{I_{R2}} = \frac{4}{1m} = 4k\Omega$
 - Also by ohms law
 - $I_{R1} = \frac{V_{CC}-V_B}{R_1} = \frac{12-4}{R_1}$ or $R_1 = \frac{8}{I_{R1}}$
 - By KCL
 - $I_{R1} = I_{R2} + I_B = 1m + \frac{I_E}{\beta+1} = 1m + \frac{1m}{101} = 1.0099mA$
 - $\Rightarrow R_1 = \frac{8}{I_{R1}} = \frac{8}{1.0099m} = 7.9216k\Omega$



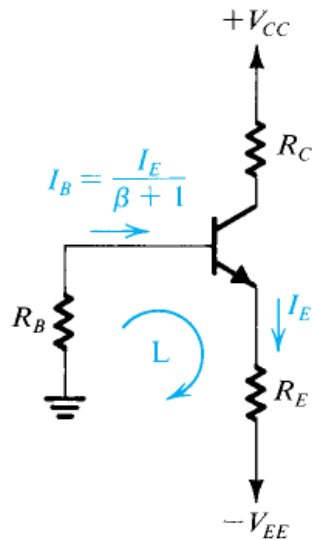
A Two-Power-Supply Version of the Classical Bias Arrangement

- if two power supplies are available, a relatively simpler biasing arrangement can be used
 - By KVL along L
 - $I_B R_B + V_{BE} + I_E R_E - V_{EE} = 0$
 - as $I_B = \frac{I_E}{\beta+1}$
 - $\frac{I_E}{\beta+1} R_B + V_{BE} + I_E R_E = V_{EE}$

$$\blacksquare I_E \left(\frac{R_B}{\beta+1} + R_E \right) = V_{EE} - V_{BE}$$

$$\blacksquare \text{ or } I_E = \frac{V_{EE}-V_{BE}}{R_E + \frac{R_B}{\beta+1}}$$

$$\blacksquare I_E = \frac{V_{EE}-V_{BE}}{R_E + \frac{R_B}{\beta+1}}$$

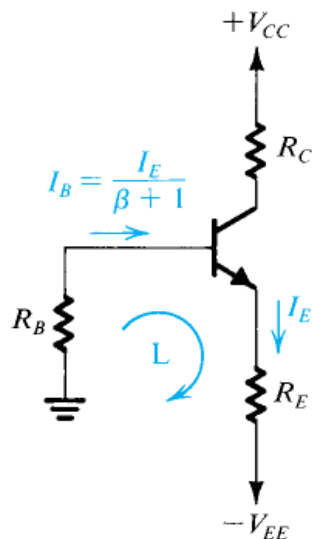


$$\bullet I_E = \frac{V_{EE}-V_{BE}}{R_E + \frac{R_B}{\beta+1}}$$

- Note that this equation is identical to the one obtained for “Classical Discrete-Circuit Bias Arrangement” (i.e. $I_E = \frac{V_{BB}-V_{BE}}{R_E + \frac{R_B}{\beta+1}}$) except for V_{EE}

replacing V_{BB} .

- Thus the two constraints mentioned previously apply here as well
- Note that if the transistor is to be used in common-base configuration, then R_B can be eliminated i.e. $R_B = 0$



Biasing Using a Collector-to-Base Feedback Resistor

- Another effective discrete-circuit biasing arrangement is possible by
 - connecting a feedback resistor R_B between the collector and the base

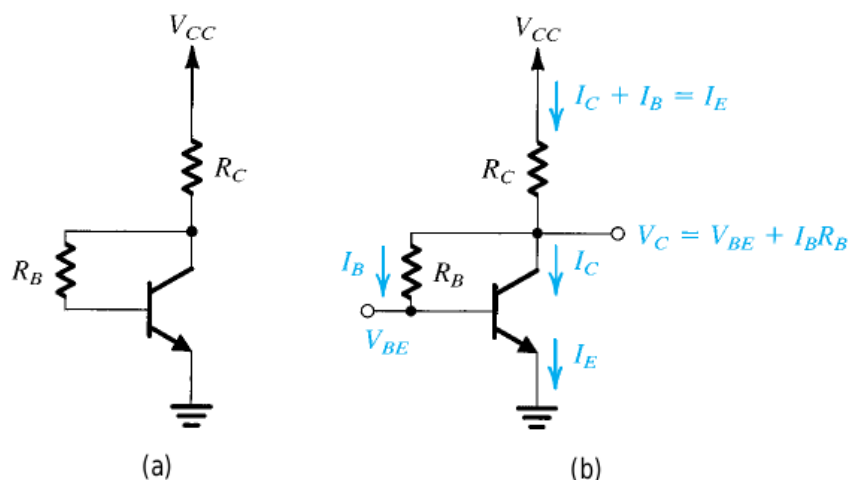


Figure 6.62 (a) A common-emitter transistor amplifier biased by a feedback resistor R_B . (b) Analysis of the circuit in (a).

- by KVL, $V_{CC} = I_E R_C + I_B R_B + V_{BE}$

- as $I_B = \frac{I_E}{\beta+1}$

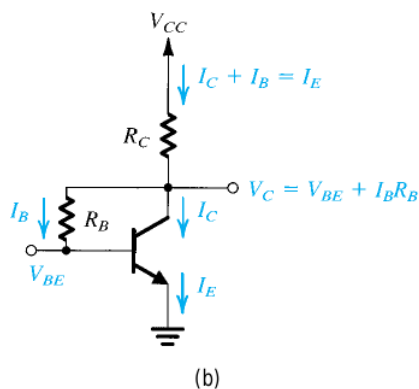
- $V_{CC} = I_E R_C + \frac{I_E}{\beta+1} R_B + V_{BE}$

- $V_{CC} - V_{BE} = I_E R_C + \frac{I_E}{\beta+1} R_B$

- $V_{CC} - V_{BE} = I_E \left(R_C + \frac{1}{\beta+1} R_B \right)$

- $\frac{V_{CC} - V_{BE}}{R_C + \frac{1}{\beta+1} R_B} = I_E$

- $I_E = \frac{V_{CC} - V_{BE}}{R_C + \frac{R_B}{\beta+1}}$



- Note that this equation is identical to the one obtained for “Classical Discrete-Circuit Bias Arrangement” (i.e. $I_E = \frac{V_{BB} - V_{BE}}{R_E + \frac{R_B}{\beta+1}}$) except for V_{CC}

replacing V_{BB} .

- thus this circuit does provide bias point stability

- i.e. to obtain stable bias point

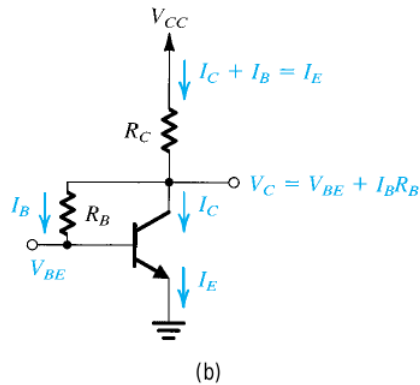
- $V_{CC} > V_{BE}$

- and $R_C > \frac{R_B}{\beta+1}$

- Also note that in this circuit arrangement, the allowable negative

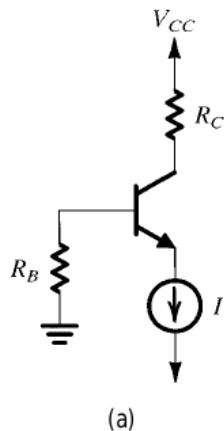
signal swing at the collector is determined by the value of R_B

- by KVL
- $V_{CB} = I_B R_B = \frac{I_E R_B}{\beta + 1}$



Biassing Using a Constant-Current Source

- Another effective way for biasing a BJT is shown in fig and it utilizes a constant-current source I
 - This circuit has the advantage that the emitter current is independent
 - of the values of β and R_B
 - thus R_B can be made large, enabling an increase in the input resistance at the base
 - without adversely affecting bias stability



- the current source I can be implemented by the circuit (known as current mirror) shown in fig b

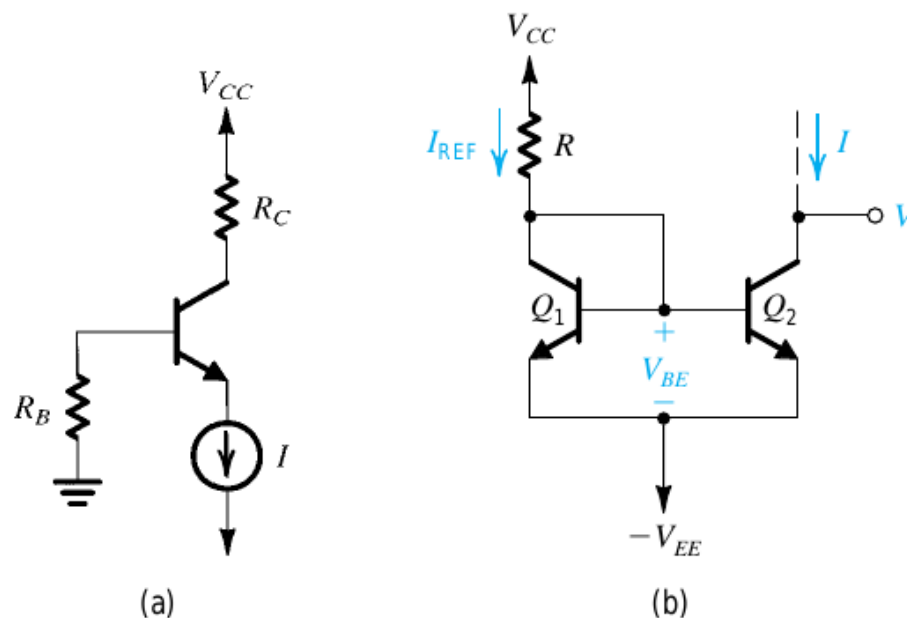
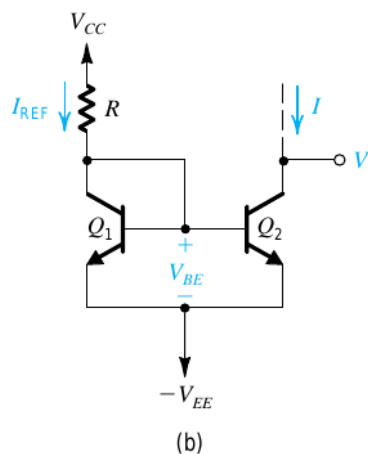


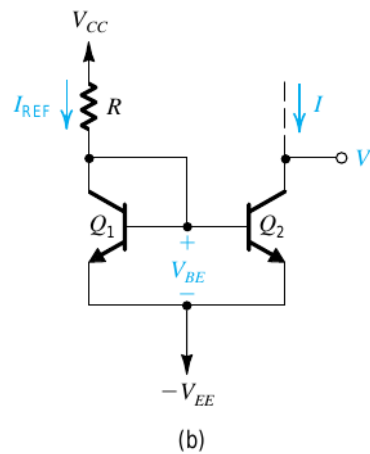
Figure 6.63 (a) A BJT biased using a constant-current source I .
 (b) Circuit for implementing the current source I .

- The circuit utilizes a pair of matched transistors Q_1 and Q_2
 - the transistor Q_1 is connected as a diode by shorting its collector to its base
 - By KVL
 - $V_{CC} = I_{REF}R + V_{BE} + (-V_{EE})$
 - $V_{CC} = I_{REF}R + V_{BE} - V_{EE}$
 - $I_{REF} = \frac{V_{CC} + V_{EE} - V_{BE}}{R}$
 - Assuming that Q_1 and Q_2 have high β values,
 - we can neglect their base currents
 - $\Rightarrow I_{REF} = I_{C,Q1}$



- As Q_1 and Q_2 have the same V_{BE}
 - $\Rightarrow$ their collector currents will be equal
 - $\Rightarrow I_{C,Q2} = I_{C,Q1} = I = I_{REF}$
 - thus $I = I_{REF} = \frac{V_{CC} + V_{EE} - V_{BE}}{R}$
 - Neglecting the Early Effect in Q_2 ,

- the collector current $I = I_{C,Q2}$ will remain constant at this value $\left(\frac{V_{CC} + V_{EE} - V_{BE}}{R} \right)$
- as long as the transistor Q_2 stays in active region.
- i.e. as long as the voltage at the collector is kept greater than that at the emitter by at least 0.3V ($V_{CE} \geq 0.3$)



Exercise 6.47

Exercise D6.48

Exercise D6.49

Exercise 6.50
